SYSTEM ARCHITECTURE

Pratibimb · hybrid physiology + ML digital twin for Type 2 Diabetes

data / control flow · evaluation only (hidden truth) · core twin / UI · component

01 DATA · SYNTHETIC ONLY

Truth simulator
richer than the twin (no inverse crime)
240 patients × 28 days, 5-min
INSPIRED subtypes SIDD · SIRD · MARD · IROD, "thin-fat" phenotype
Indian meals, late dinners, fasting
sensor lag · drift · logging errors

Synthetic EHR → FHIR R4
Bundle: Patient + ABHA 91-0000-…
Condition (SNOMED CT)
Observation (LOINC) · Medication
Statement · Device · genetics

Wearable + CGM streams
CGM every 5 min
heart rate · HRV (RMSSD)
steps · sleep stages
logged meals (73 Indian foods)
logged doses (taken / missed)

Hidden truth
plasma glucose · subtype label
never shown to the twin or the ML;
used only to score held-out patients

02 PRATIBIMB ENGINE · PYTHON 3.12

numpy · numba · scikit-learn · LightGBM

TWIN · PHYSIOLOGY

personalised from the record, re-synced every 5 minutes

EHR priors
TyG index
ADAG eAG (HbA1c)
waist/height · BMI
duration · C-peptide
TCF7L2 (weak)
→ θ prior

Twin model
extended Bergman
minimal-model ODE
G · X · I · gut · Gi
exercise · sleep · stress · 5 drug classes

EnKF sync
the live twin
64 members
states + log-params
CGM update / 5 min
4σ innovation gate
inflation 1.001

Ensemble forecast
+4 h cone
p05 · p25 · p50 · p75 · p95
habitual-meal prior
80 % intervals calibrated

What-If simulator
Indian meal swaps
post-meal walk
skip / shift a dose
short-sleep night
→ baseline vs scenario + narrative

CGM observations → EnKF update · wearables, meals, doses → model inputs

33 twin features: forecast quantiles, exceedance probabilities, posteriors

RISK LAYER · MACHINE LEARNING

Causal features
~100 per decision point, every 15 min, data ≤ t
CGM dynamics · carbs on board · meds · steps · HR/HRV · sleep · EHR
+ 33 twin features

LightGBM × 3
hyper_2h · >250 in 2 h
hypo_1h · <70 in 1 h
nocturnal_hypo · tonight
patient-level split
70 / 15 / 15

Calibrate + explain
isotonic calibration (validation patients)
TreeSHAP pred_contrib
→ plain-language drivers: "twin +2 h forecast raises risk"

Alert logic
threshold: ≤ 1 false alert / patient-day
cool-down + hysteresis
→ alert with drivers and a care note

03 SERVING · CLINICIAN

FastAPI · REST + WebSocket
GET /api/cohort · /api/patients/{id}
GET …/timeline · …/forecast
GET …/fhir (FHIR R4 Bundle)
POST …/whatif
GET /api/metrics · /cohort/validation
WS /api/stream/{id} live replay

Static demo export
pratibimb export-demo
→ JSON in web/public/demo
console runs with no backend

Clinician console
Next.js 15 · TypeScript · Tailwind v4 · SVG charts
Cohort triage, risk-sorted
Virtual patient: twin band, forecast cone, risk + TreeSHAP, alert feed
What-If Lab · EHR / FHIR sheet
Model card (reads metrics.json)

Deploy
static demo, or live api + web

Vercel · Docker Compose · uv + npm (local)

predictions

04 EVALUATION · HELD-OUT SYNTHETIC PATIENTS

→ reports/metrics.json

36 test patients never used for training or tuning · labels from hidden plasma glucose

AUROC / AUPRC + 95 % CI · ablations · Parkes grid · lead time

05 EXTERNAL CHECK · REAL OPEN DATA

→ reports/real_validation.json

CGMacros + ShanghaiT2DM · 90 records · 87 people · no tuning

Training data are synthetic (DPDP Act / HIPAA sandbox); open de-identified real CGM data only for external validation. Not a medical device.

ARCHITECTURE · COMPONENT REFERENCE

What each block is made of · and where it lives in the repo

LAYER	COMPONENT	WHAT IT DOES	TECH · CODE
DATA	**Truth simulator**	Generates 240 synthetic Indian T2D patients × 28 days from INSPIRED-style archetypes, with richer physiology than the twin, Indian lifestyle, sensor and logging artefacts.	numpy, numba · `engine/pratibimb/cohort/`
DATA	**Synthetic EHR → FHIR R4**	Labs, conditions, medications, genetics as one FHIR R4 *collection* Bundle per patient, ABHA-format identifier, LOINC and SNOMED CT codes.	`cohort/ehr.py`, `cohort/fhir.py`
TWIN	**EHR priors**	TyG index, ADAG eAG, waist/height, BMI, duration, C-peptide and TCF7L2 → prior means and spreads for S_I, S_G, β, EGP, meal rate, dawn.	`twin/priors.py`
TWIN	**Twin model**	Extended Bergman minimal model (glucose, insulin action, insulin, two gut compartments, interstitial glucose) with exercise, sleep, stress and drug effects.	numba kernels · `physiology/`
TWIN	**EnKF sync**	64-member stochastic Ensemble Kalman Filter over states + log/logit parameters; CGM update every 5 min, innovation gating, inflation.	`twin/enkf.py`, `twin/assimilate.py`
TWIN	**Forecast · What-If**	240-min ensemble forecast with a habitual-meal prior and calibrated intervals; counterfactual meals, walks, doses and sleep from the same ensemble.	`twin/forecast.py`, `api/service.py`
ML	**Features · LightGBM**	~100 causal features incl. 33 twin features at 15-min decision points; LightGBM per task (hyper_2h, hypo_1h, nocturnal_hypo).	LightGBM · `ml/features.py`, `ml/train.py`
ML	**Calibration · TreeSHAP · alerts**	Isotonic calibration on validation patients; TreeSHAP drivers mapped to clinician text; thresholds, cool-down and hysteresis.	scikit-learn · `ml/explain.py`, `ml/alerts.py`
EVALUATION	**Held-out scoring**	Patient-level 70/15/15 split; AUROC/AUPRC with bootstrap CIs, ablation, forecast RMSE/MARD vs baselines, Parkes error grid, lead time.	`ml/evaluate.py`, `ml/parkes.py` → `reports/`
SERVING	**FastAPI · static export**	REST + WebSocket replay (SPEC §5 contract), or a static JSON export so the console runs with no backend.	FastAPI, uvicorn · `api/`, `export.py`
UI	**Clinician console**	Landing page, cohort triage, virtual patient, What-If Lab, EHR/FHIR sheet, model card; offline demo mode.	Next.js 15, React 19, Tailwind v4, Radix, Motion · `web/`
DEPLOY	**Targets**	Vercel (static demo), Docker Compose (api + web), or local `uv` + `npm`.	`docker-compose.yml`, `*/Dockerfile`